

Update on SPA Paper Edits

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February 11, 2026

- **Section I:** I am working on this. Meanwhile I have attached the pdf with other edits. Since the paper's due is near, I would appreciate check on these edits while I work on this section.
- **Section III-D (Metrics):** I updated this to include privacy and security metrics. *“First, the Membership Inference Attack (MIA) AUC measures the success rate... Second, we calculate the update entropy... Higher entropy suggests that the updates contain richer, more diverse information.”*
- **Section V-B (Results Analysis):** Corrected a math inconsistency regarding the high-resource client pool. *“In our experimental setup, 10% 20% of the clients (10 out of 50) utilize high-tier adapters...”*
- **Section V-C (Perplexity):** I expanded this to be more thorough. *“This result indicates that SPA maintains superior probabilistic calibration and preserves general language modeling capabilities more effectively than competing methods.”*
- **Section V-D (Security Results):** Clarified the role of entropy in information theory. *“Having higher entropy is bad While in traditional communication systems high entropy can denote undesirable noise, in the context of federated updates, it serves as a proxy for information density.”*
- **Section V-F (Per-Class Performance):** Elaborated on the reasons for performance in intermediate categories. *“Small models often miss the tiny differences in words that separate a 'bad' review from an 'okay' one... SPA's ability to 'denoise' helps the model ignore individual human mistakes.”*
- **Section V-H (Statistical Significance):** Detailed the significance of our p-values. *“The low p-values prove that SPA's gains aren't just luck, but come from SVD filtering out the 'clash' between inconsistent client updates.”*
- **Section VI (Challenges, Limitation, and Future Work):** Combined the sections to link current limits to upcoming goals. *“Limitations Challenges, Limitation, and Future Work... Future work will investigate hierarchical projection methods and randomized SVD algorithms.”*

Note: I've also rewritten the figure captions to be much longer and more descriptive. Actual updates to the plots and graphics are still pending.

Paper Titles

1. (★) **SPA: Subspace Projection Aggregation for Heterogeneous Federated Fine-Tuning of Large Language Models**
2. (★) **HeteroLoRA: Federated Fine-Tuning of Large Language Models Across Heterogeneous Rank Constraints**
3. (★) **Rank-Adaptive Federated LoRA: A Subspace Projection Framework for Heterogeneous Edge Devices**
4. **Beyond Lowest Common Denominator: SVD-Based Rank-Adaptive Federated Learning for LLMs**
5. **Subspace-Aware Federated Adaptation: Bridging LoRA Rank Heterogeneity via Singular Value Decomposition**
6. **FedSVD: Communication-Efficient Heterogeneous Federated Fine-Tuning via Spectral Projection**
7. **Turning Device Diversity into an Asset: Subspace Projection for Heterogeneous Federated LLM Fine-Tuning**
8. **Spectral Aggregation for Privacy-Preserving Federated Fine-Tuning of Large Language Models**
9. **FedRankAdapt: Optimal Low-Rank Projection for Heterogeneous Federated LLM Adaptation**
10. **Knowledge Distillation Across Heterogeneous LoRA Ranks via Subspace Projection in Federated Learning**